ChatGPT

Building an AI Fitness Coach

Executive summary

An effective AI fitness coach should not be a single model that "figures everything out." The most robust design is a **hybrid system**: deterministic safety and clinical guardrails first, specialized scoring models for training load, recovery, adherence, and nutrition next, and an LLM last for explanation, motivation, and conversational planning. This architecture is better aligned with exercise science, privacy realities, and product risk than an LLM-only approach. Core exercise guidance should be grounded in established physical activity and resistance-training guidance, while personalization should come from user goals, preferences, constraints, and observed response over time. For healthy adults, the highest-value behaviour change is often moving from inactivity to regular training; consistency and adherence matter more than highly complex programming for most users. [1]

For this report, I assume an **open-ended budget**, **consumer-wellness first** positioning rather than immediate medical-device positioning, **mobile-first deployment** with optional web dashboards, and a broad adult population with optional segments for athletes, older adults, and rehab users. Under those assumptions, the best near-term product is a "wellness coach with escalation," not a fully autonomous rehab or medical decision system. Rehab, post-injury return, pregnancy, and disease-specific pathways should be handled through stricter screening, lower autonomy, and clinician-reviewed protocols. [2]

The data foundation should combine **self-report** with **wearable and platform data**. Self-report remains essential for goals, pain, soreness, preferences, motivation, fatigue, and perceived readiness; in sport settings, subjective self-report can be more sensitive to training response than many commonly used objective measures. Wearables add activity, sleep, heart rate, GPS, power, body metrics, and trend context, but they also introduce missingness, off-wrist artefacts, vendor-specific averaging, opaque algorithms, and variable accuracy across metrics and users. Your system should therefore model **data confidence explicitly**, not treat all inputs as equally trustworthy. [3]

For interoperability and future-proofing, use standards wherever possible: **HL7 FHIR** for health exchange, **LOINC** for observations, **SNOMED CT** for clinical terminology, and **UCUM** for units. For nutrition, start with official food composition resources such as **USDA FoodData Central**, **Health Canada's Canadian Nutrient File**, and NIH supplement databases. For physical-activity intensity mapping, the **Compendium of Physical Activities** is highly practical. For model development and benchmarking, combine official longitudinal health datasets such as **NHANES**, **All of Us**, and **UK Biobank** with domain-specific activity-recognition datasets for exercise classification and form-quality tasks. [4]

Token and cost efficiency should be designed in from day one. The core pattern is: keep a compact longitudinal user state, retrieve only the relevant slices of recent history and guidelines, store computation-heavy features outside the prompt, and use smaller on-device or distilled models where classification is enough. Retrieval-augmented generation, embedding-based memory, PEFT methods such as LoRA, distillation, caching, and optional federated personalization can all reduce cost and data exposure while maintaining quality. [5]

Coaching modalities and programming logic

Across modalities, the programming backbone is still the familiar **FITT-VP** logic: frequency, intensity, time, type, volume, and progression. Major public-health guidance converges on regular aerobic activity plus muscle-strengthening work, while resistance-training guidance adds clearer modality-specific advice for strength, hypertrophy, and power. For sport and rehab use cases, monitoring both external and internal load is important, and load changes should be individualized rather than applied as hard universal thresholds. [6]

From a product perspective, many "modalities" are really one of two things: either a **physiological objective** such as strength, hypertrophy, endurance, power, mobility, or rehab, or a **delivery format** such as group classes, home circuits, yoga, Pilates, or functional sessions. Your coach should therefore separate:
goal logic → what adaptation is desired;
program logic → what volume, intensity, and progression produce it;
experience logic → what style the user will actually do;
safety logic → what should be restricted or escalated.
That separation makes the recommendation engine much easier to personalize and much safer to audit.

Training modalities and the variables your system should optimize

Modality	Primary goal	Key variables the coach should tune	Good inputs	Important cautions	Evidence anchor
Strength	Maximal force production	Load, sets, reps, rest, exercise selection, weekly frequency, progression, effort	Training age, equipment, recent performance, soreness, pain, schedule	Heavy loading needs screening and conservative progression for novices	ACSM's 2026 update emphasizes regular participation, individualized load/volume, and heavier loading for strength goals. [7]
Hypertrophy	Muscle size	Weekly sets per muscle group, effort proximity, exercise variety, frequency, recovery	Body-part priorities, body comp goals, appetite, sleep, soreness	Over-volume is common; adherence often drops when plans become too complex	ACSM's 2026 guidance points to higher weekly volume for hypertrophy and individualization over rigid rules. [7]

Modality	Primary goal	Key variables the coach should tune	Good inputs	Important cautions	Evidence anchor
Power	Speed-strength	Moderate loads, movement velocity, low fatigue, intent, rest, phase timing	Sport, jump/ sprint profile, neuromuscular freshness	Avoid piling power work on top of excessive fatigue	ACSM's 2026 update recommends moderate loads with fast concentric intent for power. [7]
Endurance	Aerobic capacity, durability, event readiness	Zone distribution, duration, frequency, weekly load, long session timing, tapering	HR, pace, power, GPS, RPE, event calendar, sleep	Avoid abrupt load spikes and poor taper logic; use personalized baselines	WHO and HHS guidance support regular moderate/ vigorous aerobic activity; IOC load guidance supports individualized load monitoring. [8]
HIIT	Time-efficient cardiorespiratory improvement	Work:rest ratio, intensity target, session density, recovery spacing	HR response, recent load, sleep debt, RPE, user enjoyment	High-perceived difficulty hurts adherence for some users; not ideal during poor recovery or red flags	Public-health guidance supports vigorous activity; load monitoring is still needed because intensity magnifies fatigue and injury risk. [9]

Modality	Primary goal	Key variables the coach should tune	Good inputs	Important cautions	Evidence anchor
Mobility and flexibility	ROM, movement quality, symptom reduction, readiness	Frequency, duration, joint-specific focus, sequencing with strength/cardio	Pain map, stiffness, movement screen, time available	Do not treat mobility alone as rehab when pathology is present	ACSM includes flexibility/neuromotor fitness in exercise quality guidance; NCCIH notes yoga can improve strength/flexibility in some contexts. [10]
Yoga and Pilates	Mobility, balance, trunk control, recovery, mind-body adherence	Session type, intensity, duration, teacher style, scheduling	Stress, balance needs, low-back symptoms, preference for class-based practice	Benefits vary by condition and protocol; avoid over-medicalizing generic classes	NCCIH reports yoga may help flexibility, strength, and some pain conditions; Australia's 2024 Pilates evidence review shows condition-specific evidence is mixed. [11]
Rehab	Function restoration, symptom reduction, graded return	Phase, dose, tolerated ROM, pain response, exercise quality, clinician constraints	Diagnosis, red flags, pain, surgical history, clinician plan, adherence	This is where generic AI must step back and follow condition-specific pathways	WHO defines rehabilitation as interventions to optimize functioning; NICE and APTA maintain condition-specific rehab guidance. [12]

Modality	Primary goal	Key variables the coach should tune	Good inputs	Important cautions	Evidence anchor
Sport-specific	Performance transfer to a sport	Competition calendar, taper, technical demands, strength-power-endurance mix	GPS, power, HR, practice load, match calendar, coach inputs	Needs broader context than gym training alone	IOC load statements emphasize monitoring training, competition, and psychological load in athletes. [13]
Functional and group classes	High adherence through context, variety, and social format	Attendance, session substitution, intensity spread, enjoyment fit	Class type, access, social preferences, cancellation history	Treat these as delivery wrappers; underlying physiology still needs tagging	This is primarily a product-design recommendation built on standard exercise programming principles.

For most users, the AI should recommend at the **microcycle** level first: what to do this week, what to do today, and why. Meso- and macrocycles matter, but most products fail on adherence long before they fail on elegant periodization. That matches the practical thrust of ACSM's updated resistance guidance: for general adults, simply doing regular training matters more than chasing an over-optimized plan. [7]

User data, sensors, privacy, and data quality

A strong coach should ingest heterogeneous data, but it should not depend on every stream being present. In practice, you want a graded data stack: **minimum viable personalization** from onboarding and self-report, **better adaptation** from platform health data and workout logs, and **best adaptation** from longitudinal wearable, nutrition, and body-composition trends. Self-report is not a second-class source here; it is the only practical way to capture pain, mood, perceived readiness, preference, confidence, and immediate barriers such as travel or childcare. In athlete-monitoring literature, subjective wellness measures often track response to load more consistently than many physiological markers collected at rest. [14]

Platform and wearable ecosystems now expose a broad set of fitness data. **Apple HealthKit** supports activity, workouts, sleep, vitals, body measurements, and nutrition with per-type authorization. **Android Health Connect** supports activity, sleep, body metrics, nutrition, vitals, and even planned exercise sessions, with explicit permissions and user-facing rationale requirements. **Google Health API** is now the web-centric successor to Fitbit's older developer stack and is positioned as a scalable unified API for Fitbit, Pixel Watch, and compatible device data. **Garmin Health**, **Strava**, **Fitbit**, and **Oura** each provide different blends of activity, sleep, HR, HRV, GPS, power, readiness, and body metrics. [15]

Data quality is the trap. Consumer wearables are typically **better at heart rate and step count than at energy expenditure**, and performance varies by device, context, physiology, and metric. HRV is especially sensitive to measurement conditions, artefact handling, and timing. Wrist accelerometry also suffers from non-wear, placement differences, and activity misclassification; Fitbit itself documents discrepancies between intraday and daily totals because summary values can include off-wrist estimates while intraday data reflect on-wrist measurements. Nutrition logs have both random and systematic error, and self-reported intake commonly underreports energy. The correct product response is to preserve provenance, attach confidence scores, and let models reason over uncertainty rather than silently flattening everything into a single "truth." [16]

Data types, sensor sources, and the main quality problems to model

Data type	Typical source	Product value	Common quality issues	Engineering response	Sources
Goals, preferences, barriers, symptoms, pain, RPE, hunger	In-app self-report	Essential for personalization and safety	Recall bias, satisficing, skipped check-ins	Keep questions short, adaptive, and event-triggered; use confidence + missingness flags	Self-report is operationally essential; subjective wellness measures can be highly informative. [14]
Steps, minutes, workouts	HealthKit, Health Connect, Fitbit, Garmin	Baseline activity, adherence, trend detection	Source duplication, vendor summaries, off-wrist artifacts	Deduplicate by provenance and time; keep source tags	Health Connect documents deduplication for some activity/sleep aggregates; Fitbit documents off-wrist differences. [17]
HR and resting HR	Watch, band, chest strap	Recovery, intensity control, anomaly detection	Motion artefact, skin/contact issues, exercise-specific error	Prefer chest straps for training zones; down-rank noisy periods	Wearables are generally more accurate for HR than calories, but accuracy varies by context. [18]

Data type	Typical source	Product value	Common quality issues	Engineering response	Sources
HRV	Oura, Health Connect, HealthKit, some watches	Recovery trend, readiness proxy	Highly condition-sensitive, timing-sensitive, algorithm differences	Compare to personal baseline, not universal cutoffs	HRV is influenced by many factors and measurement conditions. [19]
GPS, pace, elevation	Strava, Garmin, Google Health API, device platforms	Endurance load, route context, terrain	Urban canyon noise, pause logic, smoothing differences	Use rolling aggregates and route metadata, not single-point decisions	[20]
Power and cadence	Cycling computers, smart trainers, Health Connect	Best-in-class endurance dosing for cycling/rowing	Device calibration, dropouts, incompatible manufacturers	Store calibration metadata; validate implausible spikes	Health Connect includes power records and planned sessions; Garmin/Strava expose detailed activity data. [21]
Sleep duration and stages	Oura, HealthKit, Health Connect, Fitbit, Garmin	Recovery context, behavioural coaching	Staging accuracy varies; naps and fragmented sleep differ by vendor	Use sleep mainly as trend/context signal, not a hard gate	Consumer sleep metrics vary in accuracy; use with context. [22]
Nutrition logs	In-app logs, photo logs, FoodData Central, CNF	Energy/macronutrient estimates, meal timing, habit coaching	Underreporting, portion error, reactivity, missing meals	Favor pattern coaching over “exact calorie truth”; calibrate against weight trends	NIH/NCI notes self-report dietary error is substantial. [23]

Data type	Typical source	Product value	Common quality issues	Engineering response	Sources
Body composition and weight	Smart scales, HealthKit, Health Connect, manual entry	Progress monitoring, rate-of-change safety	Hydration noise, device inconsistency, sporadic logging	Use rolling medians and expected noise bounds	Health Connect and HealthKit support body measures; treat scale data as noisy trend signals. [24]
Medical history, medications, diagnoses	Self-report, FHIR-connected clinical sources	Safety, contraindications, rehab pathways	Sensitive data, incompleteness, stale problem lists	Keep clinically important fields structured and date-stamped	FHIR, SNOMED CT, and LOINC support structured exchange and terminology. [25]

Privacy and consent should be treated as a **product feature**, not just a compliance layer. For a Canadian launch, meaningful consent under PIPEDA is the starting point, and mobile platform rules reinforce narrow, per-type permissions. HealthKit requires fine-grained authorization by data type and encrypts the HealthKit store when the device is locked; Health Connect requires explicit permissions, a user-visible rationale activity, and encourages developers to request only the data types actually needed. In the U.S., HIPAA only applies in covered-entity contexts, but the FTC's health privacy and breach-notification rules matter for many consumer health apps that are outside HIPAA. If you serve EU users, GDPR obligations around sensitive health data and cross-border processing become central. [26]

One more practical issue: **data-use terms from upstream platforms can constrain product design**. Oura's API agreement forbids using Oura API data to develop, train, fine-tune, evaluate, prompt, or otherwise provide input to AI models via the API except through Oura's own MCP/server pathways under its terms. Strava's policy similarly restricts AI-related third-party use of Strava API materials and positions Strava MCP as the authorized first-party agent-mediated interface for personal use. If your roadmap includes LLM features, platform terms are a key dependency, not a footnote. [27]

Standards, datasets, and APIs

The cleanest canonical model for this product is: **event-sourced behavioural data + standards-based health objects + a compact coaching ontology**. Use FHIR wherever clinical interoperability matters, with `Observation`, `Condition`, `MedicationStatement`, `NutritionIntake`, and physical-activity profiles for exchange; use LOINC for observation codes, SNOMED CT for clinical concepts, and UCUM for units. This makes your downstream features, analytics, and audits much easier, especially once you start mixing consumer wearables with clinical or research sources. [28]

At the exercise-content layer, there is no single definitive official “exercise API” covering technique, muscles, equipment, contraindications, and regressions/progressions. In practice, teams do better by building an internal exercise ontology seeded from the **Compendium of Physical Activities** for intensity/MET concepts, from resistance-training and rehab guidelines for safety/progression, and from clinician/coach curation for form cues and substitutions. [29]

Recommended standards, datasets, and APIs

Source	Type	What it is good for	Why it belongs in the stack	Key watch-outs	Sources
Apple HealthKit	Platform API	Activity, workouts, vitals, sleep, body, nutrition	Deep iOS integration and strong user-controlled permission model	Per-type permissions; local-store architecture affects backend sync design	[30]
Android Health Connect	Platform API	Activity, sleep, vitals, body, nutrition, planned exercise	On-device health broker for Android; good normalization point across apps	Must implement permissions/ rationale correctly; partial dedupe behaviour	[31]
Google Health API	Web API	Fitbit/Google device data with modern OAuth stack	Current strategic successor to Fitbit Web API	Migration and consent flows matter; platform evolution is active	[32]
Garmin Health API	Vendor API	Steps, HR, sleep, stress, respiration, body comp, activity data	Strong endurance/ sport signals and all-day metrics	Commercial access model; enterprise orientation	[33]
Strava API and MCP	Vendor API	Activities, routes, streams, performance history	Excellent for endurance history, GPS, pace, power, social training data	Third-party AI use is restricted; policy must be reviewed carefully	[34]
Oura API	Vendor API	Sleep, HRV, readiness, temperature-related signals	High-value recovery context	AI-use restrictions in API terms are material	[35]

Source	Type	What it is good for	Why it belongs in the stack	Key watch-outs	Sources
USDA FoodData Central	Official nutrition API/ database	Food composition, branded foods, nutrient lookup	Free, official, broad, API-ready, public-domain USDA data	Branded completeness varies; portion mapping still needs UX work	36
Canadian Nutrient File	Official nutrition database	Canadian food composition	Better fit for Canadian products and labels	Update cadence and API ergonomics are less developer-friendly than FDC	37
NIH DSLD and DSID	Official supplement resources	Supplement labels and measured ingredient estimates	Essential if you will support supplements at all	Supplement evidence and safety messaging need strict governance	38
HL7 FHIR + Physical Activity IG	Standard	Clinical interoperability	Supports exchange with EHR and other health systems	Real-world implementations vary by partner	39
LOINC + SNOMED CT + UCUM	Terminology/ units	Coding observations, concepts, and units	Necessary for clean cross-source analytics and clinical-grade exchange	Requires terminology governance, not just one-time adoption	40
Compendium of Physical Activities	Reference dataset	MET/intensity mapping and activity taxonomy	Practical bridge from free-text activities to standardized intensity	Not a personalized prescription engine by itself	41
NHANES	Official public dataset	Population modelling, nutrition/health associations	Very useful for priors, baseline distributions, and survey variables	Mostly observational; not enough for sequence personalization alone	42

Source	Type	What it is good for	Why it belongs in the stack	Key watch-outs	Sources
All of Us wearables/ Fitbit data	Official research dataset	Longitudinal wearable + survey + clinical linkage	Strong for real-world behaviour modelling and representation learning	Access and governance are research-oriented	43
UK Biobank accelerometry	Official research dataset	Large-scale activity and health-outcome modelling	Excellent for behavioural phenotyping and outcome associations	Access restrictions and cohort biases apply	44
UCI HAR + Physical Therapy Exercises	Public benchmark datasets	Activity and exercise-classification baselines	Good low-cost starting points for sensor models	Too small and too clean for final production models	45
RecoFit / exercise-recognition datasets	Public benchmark datasets	Repetition counting and gym-exercise recognition	Useful for prototype exercise-recognition models	Still need in-product data and device-specific retraining	46

AI architecture and token-efficient personalization

The most defensible architecture is a **layered hybrid**:

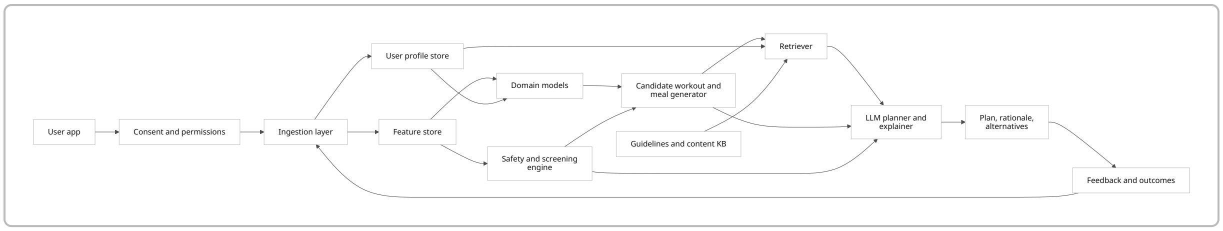
1. **Safety and eligibility engine**
   Deterministic rules for screening, contraindications, medication-related caution, pain escalation, pregnancy pathways, and rehab restrictions.

2. **Domain models**
   Specialized models for activity recognition, load/recovery scoring, adherence prediction, nutrition estimation, and ranking candidate workouts/meals.

3. **Personalization layer**
   A compact user-state representation, sequential recommender or contextual bandit logic, and retrieval of relevant history, equipment, and preferences.

4. **LLM orchestration and explanation**
   The LLM should turn structured state into plans, explanations, alternatives, and dialogue—not invent safety policy or compute primary scores.

This division is consistent with what the underlying methods are actually good at. RAG works well when the model needs just-in-time access to external context; embeddings are efficient for semantic retrieval; contextual bandits are strong for cold-start and adaptive recommendation; self-attentive sequence models are strong when you have enough history; LoRA and distillation reduce adaptation cost; federated approaches can personalize without centralizing all user data. [47]



Architecture options and where each one fits

Architecture	Best use	Strengths	Weaknesses	Recommended role	Sources
Rule engine + LLM	General wellness coaching with real safety needs	Auditable, easy to harden, easy to localize	Rules become brittle if overloaded	Baseline production architecture	Public health/ clinical guidance should sit in deterministic layers; LLM should narrate and adapt. [48]
RAG-backed LLM	Conversational planning over user history and guidelines	Keeps prompts focused; reduces need for giant context windows	Retrieval quality becomes critical	Use for explanation, summaries, plan generation	[49]
Contextual bandit	Cold start, reminder timing, message selection, content ranking	Fast online personalization and exploration/ exploitation balance	Narrower than full sequence modelling	Excellent for early-stage personalization	[50]

Architecture	Best use	Strengths	Weaknesses	Recommended role	Sources
Sequential recommender	Rich longitudinal training or meal history	Captures temporal preference and routine patterns	Needs enough data and careful offline evaluation	Use once user history is deep enough	[51]
Time-series recovery/load model	Readiness, fatigue risk, habit trend estimation	Handles rolling physiological and behavioural signals well	Easy to overfit noisy wearable data	Use for scoring, not for end-user wording	IOC load guidance + wearable-quality literature support individualized trending rather than absolute claims. [52]
Computer vision / exercise sensor model	Form check, reps, phase detection	High utility for strength/rehab UX	Data collection and labeling are expensive	Later-stage feature, not MVP blocker	[53]
Federated personalization	Privacy-sensitive on-device adaptation	Raw data can remain local	Operational complexity; privacy is improved, not magically solved	Consider for mature mobile product	[54]
Distilled / on-device models	Classification, tagging, intent, local fallback	Lower latency, lower cloud cost, better privacy	Smaller capability envelope	Ideal for edge scoring and fallback UX	[55]

Token-saving techniques that are actually worth implementing

Technique	How it saves tokens or money	Where it works best	Tradeoffs	Sources
Compact user profile	Store stable facts and rolling summaries outside the prompt	Every request	Requires careful schema/versioning	This is a design recommendation aligned with RAG/memory best practice. [56]
Retrieval-augmented generation	Inject only relevant history, guidelines, and meal/workout options	Planning and explanation	Retrieval mistakes can mislead the LLM	[49]
Embeddings over history chunks	Retrieve semantically similar prior sessions or conversations	Longitudinal dialogue and content recall	Requires chunking and freshness policy	[57]
Delta memory	Update recent changes only instead of replaying whole history	Ongoing coaching chat	Summary drift if never refreshed	Recommended product pattern
Server-side feature computation	Put features, not raw logs, into prompts	High-volume production systems	Less transparency unless provenance is shown	Recommended product pattern
Typed output schemas	Constrain response format and reduce verbose reasoning	Any structured planning call	Can feel rigid if overused	Recommended product pattern
Distillation	Smaller student models for ranking/classification	Edge and high-volume backends	Student may lose nuanced capability	[58]
LoRA or other PEFT	Cheap domain adaptation without full fine-tuning	Domain language and narrow planning styles	Still needs eval and drift monitoring	[59]

Technique	How it saves tokens or money	Where it works best	Tradeoffs	Sources
On-device inference	Avoid cloud round trips and central data exposure	Short intent, scoring, fallback personalization	Limited model size and update complexity	60
Federated learning / tuning	Learn from decentralized data without central raw-data collection	Mature mobile products	Complex MLOps; updates can still leak info if mismanaged	61

Compact user profile schema for token-efficient personalization

A compact profile should keep only the fields that materially change planning. Everything else can live in the feature store.

```
{
  "u": "user_id",
  "tz": "America/Vancouver",
  "lang": "en-CA",
  "seg": ["general_adult"],
  "g": {
    "primary": "fat_loss",
    "secondary": ["strength", "sleep"],
    "horizon_wk": 16,
    "priority": {"fat_loss": 0.6, "strength": 0.3, "sleep": 0.1}
  },
  "sched": {
    "days": ["Mon", "Tue", "Thu", "Sat"],
    "mins": {"weekday": 45, "weekend": 75},
    "access": ["home", "gym"]
  },
  "prefs": {
    "styles": ["barbell", "walks", "short_sessions"],
    "dislikes": ["burpees", "early_morning"],
    "social": "solo",
    "coach_tone": "direct_supportive"
  },
  "constraints": {
    "injuries": ["patellofemoral_pain_history"],
    "flags": [],
    "equip": ["rack", "barbell", "dumbbells", "bands"],
    "diet": ["high_protein", "no_shellfish"]
  },
```

```
  "baseline": {
    "train_age_y": 1.5,
    "steps_28d": 7400,
    "sleep_h_28d": 6.8,
    "rhr_28d": 61,
    "hrv_28d": 38
  },
  "state": {
    "adl": 0.74,
    "recov": 0.58,
    "adherence": 0.81,
    "pain": 0.1,
    "stress": 0.55
  },
  "recent": {
    "load_7d": 1.12,
    "load_28d": 1.00,
    "workouts_done_14d": 6,
    "miss_reason_top": ["travel"]
  },
  "nutrition": {
    "cal_target": 2200,
    "protein_g": 160,
    "meal_pattern": "3_plus_1",
    "log_conf": 0.62
  }
}
```

Example planning prompt templates

These are reference templates, not fixed copy.

Workout planning prompt

```
System:
You are a fitness-planning assistant. Follow safety flags exactly.
Use only the provided structured profile and retrieved evidence.
Return compact JSON plus a short user-facing explanation.

Developer:
Rules:
- Never override red flags.
- Prefer adherence over complexity.
- Keep progression conservative when recovery < 0.45 or pain > 0.2.
- Offer one main plan and one lower-friction fallback.
```

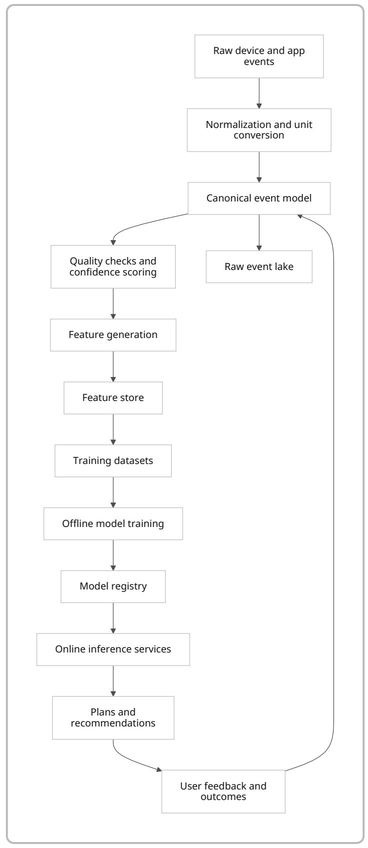
```
User state:
{{compact_profile}}

Retrieved context:
{{recent_workouts_summary}}
{{relevant_guidelines}}
{{equipment_and_constraints}}

Task:
Create today's workout for the primary goal, with warm-up, main work, cooldown,
effort targets, substitutions, and one-sentence rationale per block.
```

Nutrition coaching prompt

```
System:
You are a nutrition-coaching assistant for general wellness.
Do not diagnose or treat disease. Escalate clinically significant red flags.

Developer:
- Use Canada-first food language where possible.
- Prefer meal-pattern and nutrient-density advice when logging confidence is
low.
- If supplement questions arise, rely on official supplement resources only.

User state:
{{compact_profile}}

Retrieved context:
{{last_7d_nutrition_summary}}
{{food_database_hits}}
{{relevant_guidelines}}

Task:
Recommend today's meals or meal pattern.
State protein target distribution, fibre emphasis, hydration cue,
and one easy fallback meal for a busy day.
```

Data pipeline and model training

A practical pipeline starts with **event ingestion**, not monolithic “user snapshots.” Ingest raw events from platforms, wearables, manual logs, check-ins, and workout completions; normalize units and timestamps; map into a canonical schema; compute rolling features; and persist both raw provenance and derived features. The derived features should drive most models, while the raw events remain available for audits, debugging, and targeted retrieval. Health Connect’s data model, aggregate APIs, and data-type granularity are particularly useful examples of how to keep raw records and aggregates separate. [62]



Feature engineering should focus on **decision-useful signals**, not maximal data retention. Examples include rolling workout completion rate, days since last strength session by muscle group, 7-day versus 28-day activity change, recent sleep deviation from baseline, meal-logging confidence, soreness and pain trend, and schedule friction indicators such as travel or repeated cancellation windows. For endurance users, preserve pace/power/HR context and route metadata. For strength users, store estimated performance history by movement pattern, not just raw exercise names. For rehab users, store tolerance and symptom response by exercise family. [63]

Labeling should combine **explicit, weak, and proxy labels**:

- explicit labels: user-rated difficulty, enjoyment, pain response, goal confidence;
- weak labels: completed vs skipped, shortened vs full workout, swap frequency, meal-log completeness;
- proxy labels: changes in rolling strength estimates, pace/power trends, body-weight trend, sleep recovery trend;
- expert labels: clinician or coach review for rehab and sport-specific pathways.

For missing data, simulate the kinds of missingness you will actually see: non-wear windows, partial GPS tracks, duplicate source conflicts, meal underlogging, and stale body-weight entries. Train models to consume both values and **missingness indicators**, because missingness is often informative in a behaviour product. Non-wear and source inconsistency are well-documented issues in wearable datasets, so your simulator should reflect them deliberately. [64]

Training strategy with and without full history

When you **do not have full user history**, lean on composition of:

- onboarding goals and constraints,
- cluster-based population priors,
- contextual bandits for exploration,
- short-term self-report,
- conservative progression defaults.

This is where simple ranking models and bandits outperform overconfident sequence models. Contextual bandits were built for exactly this exploration/exploitation problem. [50]

When you **have moderate history**, add:

- personalized baselines,
- recency-weighted adherence features,
- preferred modality embeddings,
- repeated failure-mode detection,
- retrieval over recent workout and nutrition episodes.

When you **have rich longitudinal history**, sequence models become attractive for next-best-workout timing, content ranking, and behavioural forecast. Self-attentive sequential recommenders are a sensible template when histories are dense enough. [51]

For domain adaptation, use **PEFT first**. If you want an LLM to speak in your coaching style or follow your JSON schema better, LoRA-style adaptation is usually the right starting point before considering any full fine-tune. For sensitive mobile personalization at scale, federated tuning can be attractive, but it is a later-stage capability, not an MVP requirement. [65]

Evaluation, safety, governance, and UX

Evaluation should cover five dimensions at once: **efficacy, adherence, safety, personalization, and engagement**. If you optimize only for click-through, you will create an annoying coach. If you optimize only for physiological efficacy, you may create plans people abandon. If you optimize only for adherence, you may create stagnant programs. The right metric stack is hierarchical.

A good evaluation frame for this product is:

Dimension	What to measure
Efficacy	Change in strength estimates, training consistency, endurance performance proxies, body-weight/body-comp trends, sleep trend or symptom trend depending on goal
Adherence	Workout completion rate, session shortening, streak survival, time-to-dropout, logging persistence

Dimension	What to measure
Safety	Pain escalation events, red-flag triggers, unsafe recommendation rate by review, complaint/escalation rate
Personalization	Uplift versus generic plan, acceptance rate of recommended modality, satisfaction by segment, calibration of readiness predictions
Engagement	Weekly active coaching days, plan views to completion ratio, conversation helpfulness, notification opt-out rate

Where published digital-health evidence is relevant, it tends to show **modest but real average effects** for digital behaviour-change interventions on physical activity, with engagement positively associated with outcomes, though the evidence on long-term adherence remains mixed and highly implementation-dependent. That is a strong argument for measuring both behaviour change and retention directly in your own product, rather than assuming one implies the other. [66]

For experimentation, I would prioritize these A/B tests:

1. **Cold-start onboarding depth**
   Minimal onboarding versus goal-plus-constraints-plus-style onboarding.
   Outcome: first-14-day completion and plan acceptance.

2. **Static weekly plan versus adaptive daily adjustment**
   Outcome: adherence, satisfaction, and safety events.

3. **Explanation style**
   Short directive versus rationale-rich explanation.
   Outcome: completion, trust, and override rate.

4. **Reminder timing policy**
   Fixed reminders versus contextual bandit timing.
   Outcome: opens, completions, and notification fatigue.

5. **Nutrition feedback style**
   numeric/calorie-heavy versus pattern-based coaching when log confidence is low.
   Outcome: meal logging persistence and self-reported helpfulness.

6. **Fallback design**
   “You missed today” versus “2-minute salvage plan.”
   Outcome: streak survival and recovery after missed sessions.

Safety should be its own product track. At minimum, integrate evidence-based preparticipation screening such as the **CSEP Get Active Questionnaire** for Canadian users, with pregnancy-specific pathways where relevant, and align escalation logic with modern ACSM screening principles that focus on current activity, known disease, symptoms, and intended intensity rather than old-fashioned universal risk-factor counting. [67]

Your red-flag list should include, at minimum: chest pain, unexplained syncope, acute neurologic symptoms, severe shortness of breath, suspected eating disorder or rapid unsafe weight-loss behaviour, pregnancy complications, acute injury with major swelling or instability, uncontrolled hypertension or diabetes concerns if disclosed, and significant symptom worsening during rehab flows. In those states, the model should stop suggesting progression and route the user to qualified care or to a clinician-approved protocol. That is consistent with public exercise-screening and rehab guidance. [68]

On governance and regulation, the big boundary is whether you remain a **general wellness coach** or drift into **software as a medical device**. Health Canada's guidance makes clear that machine-learning-enabled software can be a medical device depending on intended use, and the FDA's mobile-medical-app framework likewise turns on function and claims. The EU AI Act adds another layer of obligations for higher-risk AI systems. If you begin diagnosing, predicting injury in a clinical sense, directing rehab for pathology, or claiming treatment or prevention outcomes, you should assume a much heavier regulatory burden. [69]

Explainability requirements are simple in spirit: always tell the user **why this plan**, **what data influenced it**, **what assumptions were made**, **how confident the system is**, and **how to override it**. Also let users manage permissions easily and see what data categories are connected. That aligns with platform expectations around user control and with broader trustworthy-AI guidance from NIST. [70]

Implementation roadmap

The most efficient build is staged. Do not start with computer vision, full meal-photo understanding, or clinical interoperability unless those are truly core to your launch thesis.

Recommended milestones, data dependencies, and exit criteria

Phase	What to build	Required data	Evaluation plan	Exit criteria
Phase one	Core coach for healthy adults: onboarding, screening, static plan generator, manual workout logging, simple nutrition coaching, rule-based safety	Self-report, basic workout logs, FoodData Central/CNF, CSEP/ACSM screening content	Offline review of recommendation quality; pilot on adherence and user satisfaction	Safe baseline plans, low unsafe recommendation rate, acceptable week-4 retention
Phase two	Wearable integration and adaptive planning: HealthKit, Health Connect, Google Health API, profile memory, recovery scoring	Platform APIs, permission flows, feature store	A/B test static vs adaptive planning; calibration by device/ source	Better adherence than static baseline without safety regression

Phase	What to build	Required data	Evaluation plan	Exit criteria
Phase three	Personalization ranking: contextual bandit for reminders and workout choice; retrieval-backed LLM explanations	Sufficient behavioural history and content embeddings	Online uplift tests; personalization fairness slices	Statistically credible uplift in completion and satisfaction
Phase four	Deeper domain models: strength progression estimator, endurance load scorer, meal pattern classifier	Rich histories, retained users, validated labels	Goal-specific efficacy metrics and clinician review where needed	Objective improvement on target goals with maintained safety
Phase five	Advanced features: sequence recommenders, on-device models, optional federated personalization, selective rehab/sport modules	Longitudinal scale, mobile MLOps, expert content governance	Segment-specific experiments and post-launch model-drift monitoring	Clear ROI versus simpler architecture

A sensible MVP dataset stack would be:

- official guidance and content sources for exercise safety and general training logic;
- FoodData Central plus Canadian Nutrient File for food composition;
- a curated internal exercise ontology;
- early in-product behavioural data for adherence learning;
- one or two public activity-recognition datasets for prototype sensor classifiers;
- NHANES or similar population data for baseline modelling only;
- later, All of Us or UK Biobank if your research and governance setup can support it. [71]

If you want the shortest path to value, the first release should optimize for four things only:

- safe eligibility and escalation,
- realistic weekly planning,
- low-friction adherence recovery when users miss sessions,
- clear explanations that feel personal without requiring giant prompts.

That will get you much closer to a useful coach than starting with flashy but brittle features.

Open questions and limitations

Some important implementation details still depend on business choices that were left open by design: target population, whether the product will remain in the general wellness category or move toward regulated clinical use, whether upstream platform terms permit your desired AI pathways for vendor data, and whether you want broad consumer breadth or narrower excellence in one segment such as strength training, running, or rehab. Those choices materially affect consent flows, data rights, recommended APIs, evaluation endpoints, and regulatory burden. [72]

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